MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW-POWER,
MULTIPLE NOR GATES, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-0038510/27A(USAF), dated 15 October 1973.

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TABLE I: Delete and substitute new table I as printed on page 2 of this amendment.

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TABLE I: Delete and substitute new table I as printed on page 3 of this amendment.

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FIGURE 4: Delete and substitute new figure 4 as printed on page 4 of this amendment.

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TABLE II, delete and substitute.

TABLE II. Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temp.	Limits Min	Limits Max	Units
Propagation delay time to low level output	t_{PHL}	V_{CC} = 5 V C_L = 50 pF R_L = 4 KΩ	4	25°C	3	66	ns
				125°C	3	99	
				-55°C	3	99	
Propagation delay time to high level output	t_{PLH}			25°C	3	60	ns
				125°C	3	90	
				-55°C	3	90	

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

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FSC 5962

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TABLE I. Electrical performance characteristics.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
1*	V_{OH}	0.7 V	0.7 V	-100 µA	4.5 V		5.5 V	5.5 V		5.5 V	5.5 V	GND	5.5 V	5.5 V		2.4		±0.12	V
2*	↕	5.5 V	5.5 V		↕	-100 µA	0.7 V	0.7 V		5.5 V	5.5 V	↕	5.5 V	5.5 V		↕		↕	↕
3*	↕	5.5 V	5.5 V				5.5 V	5.5 V	-100 µA	0.7 V	0.7 V		5.5 V	5.5 V		↕		↕	↕
4*	V_{OH}	5.5 V	5.5 V				5.5 V	5.5 V		5.5 V	5.5 V		0.7 V	0.7 V	-100 µA	2.4		±0.12	
5*	V_{OL}	GND	2.0 V	2.0 mA			GND	GND		GND	GND		GND	GND			0.3	±0.03	
6*	↕	GND	GND			2.0 mA	2.0 V	GND		↕	↕		↕	↕			↕	↕	
7*		2.0 V	↕	2.0 mA			GND	GND											
8*		GND				2.0 mA	↕	2.0 V			GND								
9*		↕						GND	2.0 mA	↕	2.0 V			GND					
10*								↕		GND	GND			2.0 V	2.0 mA				
11*									2.0 mA	2.0 V	GND		GND	GND					
12*	V_{OL}	GND	GND		4.5 V		GND	GND		GND	GND		2.0 V	GND	2.0 mA		0.3	±0.03	V
13	I_{IL}	5.5 V	0.3 V		5.5 V		5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	5.5 V		-60	-180		µA
14	↕	5.5 V	5.5 V		↕		0.3 V	5.5 V		↕	↕		↕	↕		↕	↕		↕
15		0.3 V	5.5 V				5.5 V	5.5 V			↕								
16		5.5 V	↕				↕	0.3 V			5.5 V		↕						
17		↕						5.5 V		↕	0.3 V		5.5 V						
18								↕		5.5 V	5.5 V		0.3 V	↕					
19	↕									0.3 V	5.5 V		5.5 V	5.5 V		↕			
20	I_{IL}	5.5 V	5.5 V				5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	0.3 V		-60	-180		
21*	I_{IH1}	GND	2.4 V				GND	GND		GND	GND		GND	GND			10	10X or 1 µA	
22*	↕	GND	GND				GND	2.4 V		↕	↕		↕	↕			↕	↕	
23*		2.4 V	↕				GND	GND											
24*		GND					2.4 V	↕			GND								
25*		↕					GND				2.4 V			GND					
26*							↕			GND	GND			2.4 V					
27*										2.4 V	GND		GND	GND					
28*	I_{IH1}	GND	GND		5.5 V		GND	GND		GND	GND	GND	2.4 V	GND			10	10X or 1 µA	µA

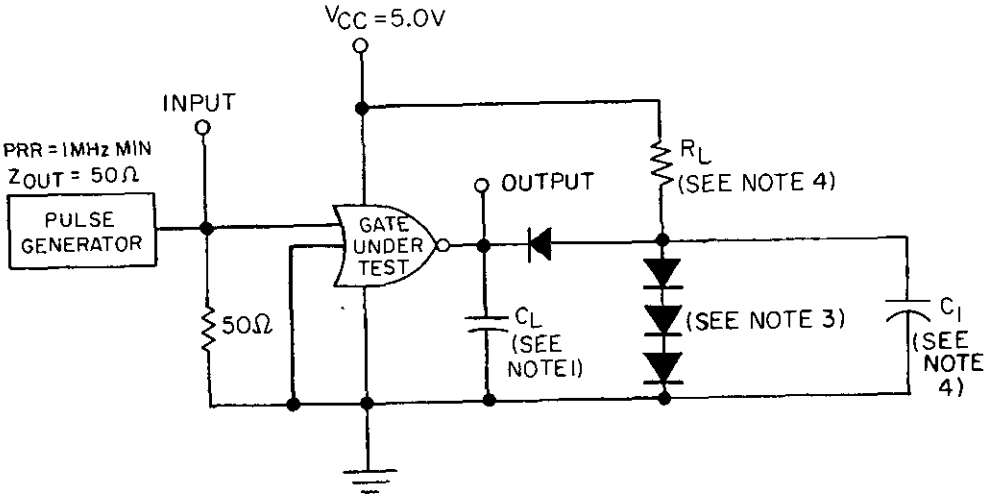
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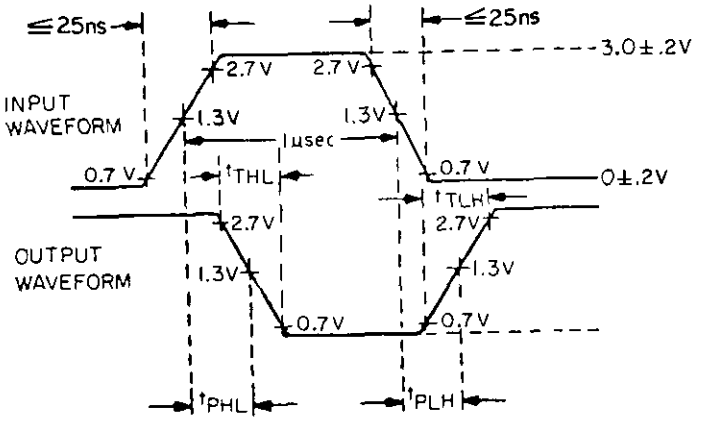
TABLE I. Electrical performance characteristics. - Continued

1/ 2/ TEST NO.	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
29	I_{IH2}	GND	5.5 V		5.5 V		GND	GND		GND	GND	GND	GND	GND			100		µA
30	↕	GND	GND		↕		GND	5.5 V		↕	↕	↕	↕	↕			↕		↕
31		5.5 V	↕				GND	GND			↕								
32		GND					5.5 V	↕			GND			↕					
33		↕					GND			↕	5.5 V			GND					
34							↕			GND	GND		↕	5.5 V					
35										5.5 V	GND		GND	GND					
36	I_{IH2}	GND	GND		-		GND	GND		GND	GND		5.5 V	GND			100		µA
37	I_{OS}	GND	GND	GND												-3	-15		mA
38	↕					GND	GND	GND								↕	↕		↕
39									GND	GND	GND								
40	I_{OS}												GND	GND	GND	-3	-15		
41	I_{CCL}	5.0 V	GND		↕		5.0 V	GND		5.0 V	GND	↕	5.0 V	GND			2.6		↕
42	I_{CCH}	GND	GND		5.5 V		GND	GND		GND	GND	GND	GND	GND			1.6		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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NOTES:
1. C_L = 50 pF minimum including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.
4. R_L = 4kΩ ±5%, C_1 = 30 pF minimum.

FIGURE 4. Switching time test circuit.

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